

Assignment-3

PCA, LDA, Linear Regression, Logistic Regression

Q.1)

A company collected the following data on employees:

Employee	Experience	Technical Score	Communication Score
A	2	65	70
B	4	80	78 78
C	5	85	82
D	6	90	88
E	8	98	95

Tasks:

1) Standardize the dataset.

Ans: Finding mean:

for experience:

$$\begin{aligned}\bar{X}_{\text{exp}} &= \frac{2+4+5+6+8}{5} \\ &= \frac{25}{5} \\ &= 5\end{aligned}$$

for technical score:

$$\begin{aligned}\bar{X}_{\text{tec}} &= \frac{65+80+85+90+98}{5} \\ &= \frac{418}{5} \\ &= 83.6\end{aligned}$$

for communication score:

$$\begin{aligned}\bar{X}_{\text{com}} &= \frac{70+78+82+88+95}{5} \\ &= \frac{413}{5} \\ &= 82.6\end{aligned}$$

Now, finding standard deviation:

for experience:

X	$d = X - \bar{X}_{\text{exp}}$	d^2
2	-3	9
4	-1	1
5	0	0
6	1	1
8	3	9
	$\Sigma d = 0$	$\Sigma d^2 = 20$

Now,

$$\text{Variance} = \frac{\sum d^2}{n-1} \quad (\text{for sample data})$$

$$= \frac{20}{5-1}$$

$$= 5$$

So, Standard deviation (σ) = $\sqrt{\text{variance}}$

$$= \sqrt{5}$$

$$= 2.236$$

For technical score:-

X	$d = X - \bar{X}_{\text{tec}}$	d^2
65	$65 - 83.6 = -18.6$	345.96
80	$80 - 83.6 = -3.6$	12.96
85	$85 - 83.6 = 1.4$	1.96
90	$90 - 83.6 = 6.4$	40.96
98	$98 - 83.6 = 14.4$	207.36

$$\sum d^2 = 609.2$$

Now,

$$\text{variance} = \frac{\sum d^2}{n-1}$$

$$= \frac{609.2}{5-1}$$

$$= \frac{609.2}{4}$$

$$= 152.3$$

So, Standard deviation (σ) = $\sqrt{\text{variance}}$

$$= \sqrt{152.3}$$

$$= 12.34$$

for Communication Score:

X	$d = X - \bar{X}_{\text{com}}$	d^2
70	$70 - 82.6 = -12.6$	158.76
78	$78 - 82.6 = -4.6$	21.16
82	$82 - 82.6 = -0.6$	0.36
88	$88 - 82.6 = 5.4$	29.16
95	$95 - 82.6 = 12.4$	153.76
		$\sum d^2 = 363.2$

Now,

$$\begin{aligned}\text{Variance} &= \frac{\sum d^2}{n-1} \\ &= \frac{363.2}{5-1} \\ &= \frac{563.2}{4} \\ &= 90.8\end{aligned}$$

$$\begin{aligned}\text{So, standard deviation } (\sigma) &= \sqrt{\text{Variance}} \\ &= \sqrt{90.8} \\ &= 9.53\end{aligned}$$

Finally, chart for standardized values of employees:

Employee	Standardized experience	Standardized technical score	Standardized Communication score
A	$(2-5)/2.236 = -1.34$	-1.507	-1.322
B	$(4-5)/2.236 = -0.447$	-0.292	-0.483
C	$(5-5)/2.236 = 0$	0.113	-0.063
D	$(6-5)/2.236 = 0.447$	0.519	0.567
E	$(8-5)/2.236 = 1.342$	1.167	1.301

Task 2:- Compute the covariance matrix.

Soln:

Here, standardized matrix is

$$Z = \begin{bmatrix} -1.34 & -1.507 & -1.322 \\ -0.447 & -0.292 & -0.483 \\ 0 & 0.113 & -0.063 \\ 0.447 & 0.519 & 0.567 \\ 1.342 & 1.167 & 1.301 \end{bmatrix}$$

$$\therefore Z^T = \begin{bmatrix} 1.34 & -0.447 & 0 & 0.447 & 1.342 \\ -1.507 & -0.292 & 0.113 & 0.519 & 1.167 \\ -1.322 & -0.483 & -0.063 & 0.567 & 1.301 \end{bmatrix}$$

Now, Covariance matrix,

$$C = Z^T Z$$

$$= \begin{bmatrix} -1.34 & -0.447 & 0 & 0.447 & 1.342 \\ -1.507 & -0.292 & 0.113 & 0.519 & 1.167 \\ -1.322 & -0.483 & -0.063 & 0.567 & 1.361 \end{bmatrix}$$

$$\begin{bmatrix} -1.34 & -1.507 & -1.322 \\ -0.447 & -0.292 & -0.483 \\ 0 & 0.113 & -0.063 \\ 0.447 & 0.519 & 0.567 \\ 1.342 & 1.167 & 1.361 \end{bmatrix}$$

$$\therefore C = \begin{bmatrix} 1 & 0.986 & 0.990 \\ 0.986 & 1 & 0.992 \\ 0.990 & 0.992 & 1 \end{bmatrix}$$

Task 3: Determine eigenvalues and eigenvectors.

$$\det(C - \lambda I) = 0$$

$$\text{or } \begin{vmatrix} 1-\lambda & 0.986 & 0.990 \\ 0.986 & 1-\lambda & 0.992 \\ 0.990 & 0.992 & 1-\lambda \end{vmatrix} = 0$$

$$\text{or } (1-\lambda) \begin{vmatrix} 1-\lambda & 0.992 \\ 0.992 & 1-\lambda \end{vmatrix} - 0.986 \begin{vmatrix} 0.986 & 0.992 \\ 0.990 & 1-\lambda \end{vmatrix} + 0.990 \begin{vmatrix} 0.986 & 1-\lambda \\ 0.990 & 0.992 \end{vmatrix} = 0$$

This way, it will take too much time.

So, I will rather solve using characteristic equation.

for 3x3 matrix, the characteristic equation is

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

Here,

$$S_1 = 1 + 1 + 1 = 3$$

$$S_2 = \text{1st principal minor} + \text{2nd principal minor} + \text{3rd principal minor}$$

Here,

$$\begin{aligned} \text{1st principal minor} &= \begin{vmatrix} 1 & 0.986 \\ 0.986 & 1 \end{vmatrix} \\ &= 1 - 0.972 \\ &= 0.0278 \end{aligned}$$

$$\begin{aligned} \text{2nd principal minor} &= \begin{vmatrix} 0.986 & 1 & 0.990 \\ 0.990 & 1 & 1 \end{vmatrix} \\ &= 1 - (0.999)^2 \\ &= 1 - 0.998010 \\ &= 0.0199 \end{aligned}$$

$$\begin{aligned} \text{3rd principal minor} &= \begin{vmatrix} 1 & 0.992 \\ 0.992 & 1 \end{vmatrix} \\ &= 1 - (0.992)^2 \\ &= 1 - 0.984064 \\ &= 0.015936 \end{aligned}$$

$$\begin{aligned} \therefore S_2 &= 0.0278 + 0.0199 + 0.015936 \\ &= 0.06364 \end{aligned}$$

Again,

$$S_3 = \det(C)$$

$$\begin{aligned} &= \begin{vmatrix} 1 & 0.986 & 0.990 \\ 0.986 & 1 & 0.992 \\ 0.990 & 0.992 & 1 \end{vmatrix} \\ &= 1 \begin{vmatrix} 1 & 0.992 \\ 0.992 & 1 \end{vmatrix} - 0.986 \begin{vmatrix} 0.986 & 0.992 \\ 0.990 & 1 \end{vmatrix} \\ &\quad + 0.990 \begin{vmatrix} 0.986 & 1 \\ 0.990 & 0.992 \end{vmatrix} \end{aligned}$$

$$= 1(1 - 0.984064) - 0.986(0.986 - 0.98208) + 0.990(0.978112 - 0.990)$$

$$= 0.015936 - 0.986 \times 0.00392 + 0.990 \times (-0.011888)$$

$$= 0.015936 - 0.00392 - 0.01176912$$

$$\therefore S_3 = 0.00030176$$

So, characteristic equation is

$$\lambda^3 - 3\lambda^2 + 0.06364\lambda - 0.00030176 = 0$$

Solving; [I used cubic equation solver from google]

$$\lambda = 2.9786$$

$$\lambda = 0.01410$$

$$\lambda = 0.00714$$

Here,

variance covered by $\lambda = 2.9786$ is

$$= \frac{2.9786}{2.9786 + 0.1419 + 0.00714}$$

$$= 99.28\%$$

So, $\lambda = 2.9786$ is the principal component analysis and captures almost all the variance of the data.

So, finding eigen value vectors it only;

Finding eigen vectors along principal component $\lambda = 2.9786$
 we have $A [C\lambda = \lambda v]$

$$\text{or } (C - \lambda I) v = 0$$

$$\text{or } \left[\begin{pmatrix} 1 & 0.986 & 0.990 \\ 0.986 & 1 & 0.992 \\ 0.990 & 0.992 & 1 \end{pmatrix} - 2.9786 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \right] v = 0$$

$$\text{or } \begin{pmatrix} 1 - 2.9786 & 0.986 - 0 & 0.990 - 0 \\ 0.986 - 0 & 1 - 2.986 & 0.992 - 0 \\ 0.990 - 0 & 0.992 - 0 & 1 - 2.9786 \end{pmatrix} v = 0$$

$$\text{or } \begin{pmatrix} -1.9786 & 0.986 & 0.990 \\ 0.986 & -1.986 & 0.992 \\ 0.990 & 0.992 & -1.9786 \end{pmatrix} v = 0$$

$$\text{or } \begin{pmatrix} -1.9786 & 0.986 & 0.99 \\ 0.986 & -1.986 & 0.992 \\ 0.990 & 0.992 & -1.9786 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$$

where $v = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$

$$\text{or } -1.9786x_1 + 0.986x_2 + 0.99x_3 = 0 \quad \text{--- (i)}$$

$$\text{or } 0.986x_1 - 1.986x_2 + 0.992x_3 = 0 \quad \text{--- (ii)}$$

$$\text{or } 0.990x_1 + 0.992x_2 - 1.9786x_3 = 0 \quad \text{--- (iii)}$$

Solving (i), (ii) and (iii), we have

$$x_1 = 0.5777$$

$$x_2 = 0.578$$

$$x_3 = 0.577$$

So, eigenvector corresponding to $\lambda = 2.9786$, is

$$v_1 = \begin{pmatrix} 0.5777 \\ 0.578 \\ 0.577 \end{pmatrix}$$

Similarly,

$$v_2 = \begin{pmatrix} 0.707 \\ -0.707 \\ 0 \end{pmatrix} \text{ and}$$

$$v_3 = \begin{pmatrix} 0.408 \\ 0.408 \\ -0.816 \end{pmatrix}$$

4. Compute explained variance ratio

Ans:

Here, $\lambda_1 = 2.98$, $\lambda_2 = 0.018$ and $\lambda_3 = 0.002$

So,

$$\text{total variance} = 2.98 + 0.018 + 0.002 \\ = 3$$

$$\text{PC1: } \frac{2.98}{3} \times 100\% \\ = 99.3\%$$

$$\text{PC2: } \frac{0.018}{3} \times 100\% \\ = 0.6\%$$

$$\text{PC3: } \frac{0.002}{3} \times 100\% \\ = 0.1\%$$

5. Reduce the data to two principal components.

Soln:

Here,

$$C = \text{standardized matrix} = \begin{bmatrix} -1.342 & -1.507 & -1.322 \\ -0.447 & -0.292 & -0.483 \\ 0 & 0.113 & -0.063 \\ 0.447 & 0.519 & 0.567 \\ 1.342 & 1.167 & 1.301 \end{bmatrix}$$

$$W = \text{eigenvector matrix} = \begin{bmatrix} 0.577 & 0.707 & 0.408 \\ 0.578 & -0.707 & 0.408 \\ 0.577 & 0 & -0.816 \end{bmatrix}$$

$$\text{Now, Using PC1 and PC2} = \begin{bmatrix} 0.577 & 0.707 \\ 0.578 & -0.707 \\ 0.577 & 0 \end{bmatrix}$$

transformed data

$$= CW$$

$$= \begin{bmatrix} -1.342 & -1.507 & -1.322 \\ -0.447 & -0.292 & -0.483 \\ 0 & 0.113 & -0.063 \\ 0.447 & 0.519 & 0.567 \\ 1.342 & 1.167 & 1.301 \end{bmatrix} \begin{bmatrix} 0.577 & 0.707 \\ 0.578 & -0.707 \\ 0.577 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} -2.40 & -0.12 \\ -0.70 & -0.11 \\ 0.03 & -0.08 \\ 0.89 & -0.05 \\ 2.18 & 0.36 \end{bmatrix}$$

$\Rightarrow$ These two columns are the new dataset.

6. Discuss information loss due to dimensionality reduction.

Ans: Original dimensions is 3.

Reduced dimensions is 2.

Variance retained = $99.3 + 0.6 = 99.9\%$

Information lost = $100 - 99.9 = 0.1\%$

So, almost no useful information is lost.

7. State one advantage and one limitation of PCA.

Ans: Advantage of PCA:

$\rightarrow$ PCA reduces the number of features while preserving most of the information making models faster and reducing dimensionality.

Limitation of PCA:

$\rightarrow$ PCA is unsupervised and does not consider class labels, so the directions of maximum variance may not be the most useful for prediction or classification.

Question 2: Linear Discriminant Analysis (LDA):

Healthy class: (2,3), (3,4), (4,3), (3,2)

Diseased class: (7,8), (8,9), (9,8), (8,7)

Tasks:

1. Compute class mean.

Ans: $\mu_{1x} = \frac{2+3+4+3}{4} = 3$

$$\mu_{1y} = \frac{3+4+3+2}{4} = 3$$

$$\therefore \mu_1 = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

Again, $\mu_{2x} = \frac{7+8+9+8}{4} = 8$

$$\mu_{2y} = \frac{8+9+8+7}{4} = 8$$

$$\therefore \mu_2 = \begin{bmatrix} 8 \\ 8 \end{bmatrix}$$

2. Compute within-class scatter matrix.

Ans: The within-class scatter matrix is

$$S_W = S_1 + S_2$$

$$S_1 = \sum (x - \mu_1) (x - \mu_1)^T$$

$$(x - \mu_1) = \begin{bmatrix} 2-3 & 3-3 \\ 3-3 & 4-3 \\ 4-3 & 3-3 \\ 3-3 & 2-3 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$(x - \mu_1)^T = \begin{bmatrix} -1 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix}$$

Similarly, for S_2 :

$$S_2 = \sum (x - \mu_2) (x - \mu_2)^T$$

4x2 2x4

$$x - \mu_2 = \begin{bmatrix} 7-8 & 8-8 \\ 8-8 & 9-8 \\ 9-8 & 8-8 \\ 8-8 & 7-8 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$(x - \mu_2)^T = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix}$$

Now,

$$S_1 = \sum (x - \mu_1)(x - \mu_1)^T$$

$$= \begin{bmatrix} -1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix}$$

$$S_1 = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

Similarly; $S_2 = \sum (x - \mu_2)(x - \mu_2)^T$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\therefore S_2 = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}$$

Final within-class scatter matrix:

$$S_w = S_1 + S_2$$

$$= \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} + \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}$$

3. Compute between-class scatter matrix.

Ans!

$$\text{Here, } S_W = \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}, \mu_1 = \begin{bmatrix} 3 \\ 3 \end{bmatrix}, \mu_2 = \begin{bmatrix} 8 \\ 8 \end{bmatrix}$$

Between-class scatter matrix

$$S_B = (\mu_1 - \mu_2)(\mu_1 - \mu_2)^T$$

$$\text{Here, } \mu_1 - \mu_2 = \begin{bmatrix} 3 \\ 3 \end{bmatrix} - \begin{bmatrix} 8 \\ 8 \end{bmatrix} = \begin{bmatrix} -5 \\ -5 \end{bmatrix}$$

$$[\mu_1 - \mu_2]^T = [-5 \quad -5]$$

$$\text{Now, } S_B = (\mu_1 - \mu_2)(\mu_1 - \mu_2)^T = \begin{bmatrix} -5 \\ -5 \end{bmatrix} [-5 \quad -5]$$

$$\therefore S_B = \begin{bmatrix} 25 & 25 \\ 25 & 25 \end{bmatrix}$$

4. Finding S_W^{-1} :-

$$S_W^{-1} = \frac{1}{16} \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}$$

$$\therefore S_W^{-1} = \begin{bmatrix} 0.25 & 0 \\ 0 & 0.25 \end{bmatrix}$$

Now,

$$\begin{aligned} S_W^{-1} S_B &= \begin{bmatrix} 0.25 & 0 \\ 0 & 0.25 \end{bmatrix} \begin{bmatrix} 25 & 25 \\ 25 & 25 \end{bmatrix} \\ &= \begin{bmatrix} 6.25 & 6.25 \\ 6.25 & 6.25 \end{bmatrix} \end{aligned}$$

Computing the projection vector (w) = $S_W^{-1} S_B (\mu_2 - \mu_1)$

$$\begin{aligned} &= \begin{bmatrix} 0.25 & 0 \\ 0 & 0.25 \end{bmatrix} \begin{bmatrix} 5 \\ 5 \end{bmatrix} \\ &= \begin{bmatrix} 1.25 \\ 1.25 \end{bmatrix} \end{aligned}$$

$$\|w\| = \sqrt{(1.25)^2 + (1.25)^2}$$

$$= 1.7678$$

$$w = \begin{pmatrix} \frac{1.25}{1.7678} \\ \frac{1.25}{1.7678} \end{pmatrix} = \begin{pmatrix} 0.7071 \\ 0.7071 \end{pmatrix}$$

So, the optimal LDA projection vector $(w) = \begin{pmatrix} 0.7071 \\ 0.7071 \end{pmatrix}$

5. Project all samples onto LDA axis.

Ans:

we have $w = \begin{pmatrix} 0.7071 \\ 0.7071 \end{pmatrix}$

To project each sample onto the LDA axis,
 $y = w^T x$

For Healthy class:

$$y = \begin{pmatrix} 0.7071 & 0.7071 \end{pmatrix}^T = (0.7071 \ 0.7071)$$

Sample 1:

$$y = [0.7071 \ 0.7071] \begin{bmatrix} 2 \\ 3 \end{bmatrix} = 1.4142 + 2.1213 = 3.5355$$

Sample 2:

$$y = [0.7071 \ 0.7071] \begin{bmatrix} 3 \\ 4 \end{bmatrix} = 4.9497$$

Sample 3:

$$y = [0.7071 \ 0.7071] \begin{bmatrix} 4 \\ 5 \end{bmatrix} = 4.9497$$

Sample 4:

$$y = [0.7071 \ 0.7071] \begin{bmatrix} 3 \\ 2 \end{bmatrix} = 3.5355$$

For Diseased class:

Sample 1:

$$y = [0.7071 \ 0.7071] \begin{bmatrix} 7 \\ 8 \end{bmatrix} = 10.6065$$

Sample 2:

$$y = [0.7071 \ 0.7071] \begin{bmatrix} 8 \\ 9 \end{bmatrix} = 12.0207$$

Sample 3:

$$y = [0.7071 \ 0.7071] \begin{bmatrix} 9 \\ 8 \end{bmatrix} = 12.0207$$

Sample 4:

$$y = [0.7071 \ 0.7071] \begin{bmatrix} 8 \\ 7 \end{bmatrix} = 10.6065$$

6. Discuss linear separability.

Ans:

Healthy class occupies the interval:

$$3.5355 \leq y \leq 4.9497$$

Diseased class occupies the interval:

$$10.6065 \leq y \leq 12.0207$$

∴ There is a gap between 4.9497 and 10.6065

Hence,

$$4.9497 < 10.6065$$

The two intervals do not overlap.

Since there is no overlap between the projected values of the two classes, they can be separated by a single threshold.

∴ The Healthy and Diseased classes are perfectly linearly separable after LDA projection.

* Compare PCA and LDA.

Ans:

PCA	LDA
① Unsupervised learning technique.	① Supervised learning technique.
② Does not use class labels.	② Use class labels.
③ Maximizes total variance.	③ Maximizes class separability.
④ Finds directions with maximum variance.	④ Finds directions with maximum discrimination.

Question 3: Linear Regression

House dataset:

House size
(sq. ft.)

Price (NPR Lakhs)

600

45

800

58

1000

72

1200

85

1400

96

1600

110

Tasks:

1. Derive Ordinary Least Squares solution.

Soln:

Linear regression assumes $\hat{y} = b_0 + b_1 x$ where

b_0 = intercept

b_1 = slope

The OLS objective is to minimize n

$$J(b_0, b_1) = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

2. Compute slope and intercept!

Here,

$$\text{Avg. house size } (\bar{x}) = \frac{600 + 800 + 1000 + 1200 + 1400 + 1600}{6}$$

$$= 1100$$

$$\text{Avg. house price } (\bar{y}) = \frac{45 + 58 + 72 + 85 + 96 + 110}{6}$$

$$= 77.67$$

$$\text{Now, } b_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

$$= \frac{45200}{700000}$$

$$\therefore b_1 = 0.0645714$$

$$b_0 = \bar{y} - b_1 \bar{x}$$

$$= 77.6667 - 0.0645 \times 1100$$

$$\therefore b_0 = 6.638$$

$$\text{So, } \hat{y} = 6.638 + 0.06456x$$

3. Predict price for 1500 sq.ft

Ans: we have

$$\hat{y} = 6.638 + 0.06456x$$

$$x = 1500$$

$$\therefore \hat{y} = 6.638 + 0.06456 \times 1500$$

$$= 6.638 + 96.857$$

$$\hat{y} = 103.50 \text{ lakhs}$$

4. Compute Mean Squared Error (MSE).

Ans:

Here,	House size	Actual	Predicted	Error ²
	600	45	45.381	0.145
	800	58	58.295	0.087
	1000	72	71.210	0.624
	1200	85	84.128	0.768
	1400	96	97.038	1.078
	1600	110	109.952	0.062

$$\text{Total Squared Error} = 2.705$$

$$2.705$$

$$\text{Mean Squared Error} = \frac{2.705}{6}$$

$$= 0.451$$

5. Compute R^2 score

$$\text{Ans: } R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}$$

$$SS_{\text{res}} = \sum (y_i - \hat{y}_i)^2$$

$$SS_{\text{tot}} = \sum (y_i - \bar{y})^2$$

Using the given dataset

$$R^2 = 0.9991$$

6. Comment on model performance.

Ans: Here, MSE and R^2 score values are 0.451 and 0.9991 respectively. The value of MSE 0.451 means the predicted prices are very close to the actual prices. Similarly, The value of R^2 is 0.9991, means 99.91% of the variation in house prices is explained by house size.

Question 4: Logistic Regression

Model:

$$P(Y=1) = \frac{1}{1 + \exp(-(-5 + 0.06x_1 + 0.08x_2))}$$

where:

x_1 = Annual income (NPR lakhs)

x_2 = Existing loan amount (NPR lakhs)

customer:

Income = 80 lakhs

Loan = 25 lakhs

Tasks:

1. Compute linear score z

Ans: Here,

$$Z = -5 + 0.06 \times 80 + 0.08 \times 25$$
$$= -5 + 4.8 + 2$$

$$\boxed{Z = 1.8}$$

2. Compute probability of default.

Ans: Here,

The sigmoid function is

$$P(Y=1) = \frac{1}{1 + e^{-z}}$$

$$z = 1.8$$

So,

$$P = \frac{1}{1 + e^{-1.8}}$$

$$= \frac{1}{1 + 0.1653}$$

$$= \frac{1}{1.1653}$$

$\therefore P = 0.8581$

$\therefore$ Probability of default = 85.81%

3. Predict class using thresholds 0.5 and 0.7.

Ans:

Since $P = 0.8581$
 If threshold = 0.5
 $0.8581 > 0.5$

Prediction $\rightarrow$ class 1

Decision rule:

If $P \geq$ threshold
 $\rightarrow$ class 1.

Otherwise
 $\rightarrow$ class 0.

Threshold = 0.7

$$0.8581 > 0.7$$

Prediction $\rightarrow$ class 1

4. Effect of threshold of FP and FN.

Ans:

decreases

Increasing threshold increases false positives.

Increasing threshold increases false negatives.

5. Why cross-entropy loss is used instead of MSE.

Ans:

Cross-Entropy Loss is preferred because it is specifically designed for classification problems, provides larger penalties for incorrect predictions, converges faster, and is based on maximum likelihood estimation. MSE is mainly suitable for regression.

6. State two assumptions of Logistic Regression.

Ans:

Assumption 1: The dependent variable is binary.

Assumption 2: Observations are independent of each other.